

Required Test Points

Power

- TP8 / +BATT
- TP47 / +5V_IN
- TP13 / TP26 / GND

ADC

- TP52 / +VIN
- TP30 / +VBCKP
- TP7 / VBAT_MEAS
- TP11 / +3.3V
- TP24 / +3.3V_GPS

GPIO

- TP50 / HARD_RESET
- TP49 / CHRG_DET
- TP12 / UVP_N
- TP1 / 3V3_PSM

UART1

- DUT.RX – J6-pin1 / nRF9160 debug RX (data into nRF91)
- DUT.TX – J6-pin2 / nRF9160 debug TX (data out from nRF91)

UART2

- DUT.RX – J8-pin1 / nRF52840 debug RX (data into nRF52)
- DUT.TX – J8-pin2 / nRF52840 debug TX (data out from nRF52)

I2C

- SCL – TP17
- SDA – TP18

SPI

- N/A

Electrical Power Test

Objectives

- Ensure board can receive power at battery input terminal.
 - Ensure 5V load sharing works.
1. Apply +2.5V to +BATT test point.
 2. Ensure device electrical state.
 - a. Ensure +VIN test point voltage is below 0.3V.
 - b. Ensure +VBCKP test point voltage is below 0.3V.
 - c. Ensure UVP_N test point voltage is digital low.
 - d. Ensure near-zero current consumption.
 3. Apply +3.2V to +BATT test point.
 4. Ensure device electrical state.
 - a. Ensure +VIN test point voltage is same as +BATT.
 - b. Ensure regulated +3.3V test point voltage is within 3.2V – 3.4V.
 - c. Ensure +VBCKP test point voltage is within +2.4V – 2.6V.
 - d. Ensure VBAT_MEAS is correct voltage.
 - e. Ensure UVP_N test point voltage is digital high.
 - f. Ensure proper current consumption (no short circuits)
 5. Apply +3.6V to +BATT test point.
 6. Ensure device electrical state.
 - a. Ensure +VIN test point voltage is same as +BATT.
 7. Assert HARD_RESET test point to digital high.
 8. Ensure device electrical state.
 - a. Ensure +VIN test point voltage is below 0.3V.
 - b. Ensure UVP_N test point voltage is digital low.
 9. Apply +5V to +5V_IN test point.
 10. Ensure device electrical state.
 - a. Ensure +VIN test point voltage is 5V.
 - b. Ensure regulated +3.3V test point voltage is within 3.2V – 3.4V.
 - c. Ensure +VBCKP test point voltage is within +2.4V – 2.6V.
 - d. Ensure UVP_N test point voltage is digital high.
 - e. Ensure CHRG_DET test point is digital low.
 11. Set HARD_RESET test point to digital low.
 12. Remove +5V.

Flash Manufacturing Firmware

Objectives

- Ensure both microcontrollers are programmed with correct firmware for testing.
1. Flash nRF9160 modem firmware 1.3.5.
 2. In parallel, flash nRF52840 with manufacturing firmware.
 3. When modem firmware is complete, flash nRF9160 with manufacturing firmware.

Execute Test Firmware

Objectives

- Communicate via UART with nRF9160 manufacturing firmware to execute various tests.
1. Ensure the board responds to NOP command.
 2. Query board for its VBAT_MEAS reading.
 3. Ensure VBAT_MEAS reading is appropriate for the VBAT voltage.
 4. Send POST command.
 5. Ensure POST pass criteria.
 - a. Verify correct chip id for all sensors, ICs, and modems.
 - b. Read, modify, write, read test for external flash.
 6. Read IMEI and all ICCIDs.
 7. Read accelerometer X/Y/Z and temperature values.
 - a. Ensure values are correct by comparing them to reference values from MTIB.
 8. Read altimeter pressure and temperature values.
 - a. Ensure values are correct by comparing them to reference values from MTIB.
 9. Ensure 3.3V_GPS voltage is between 3.2 and 3.4V.
 10. Send command to turn off GPS power.
 11. Ensure 3.3V_GPS voltage is <0.3V.

Personalize Device

Objectives

- Personalize device via UART.
1. If test firmware passed, personalize the device with data obtained from CoreCloud.